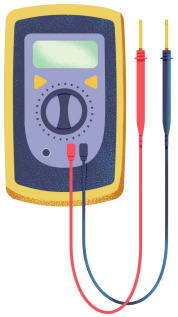
SNC1W - GRADE 9
SCIENCE

CONDUCTORS AND INSULATORS: REVIEW

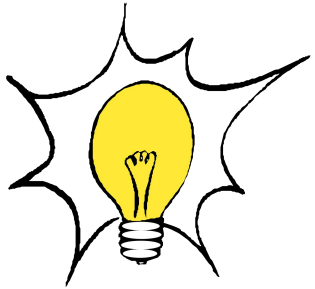


Learning Goal

Today we will:

- Determine which materials **conduct** or **insulate** by using evidence from our experiment of how they transfer electric charge

Success Criteria



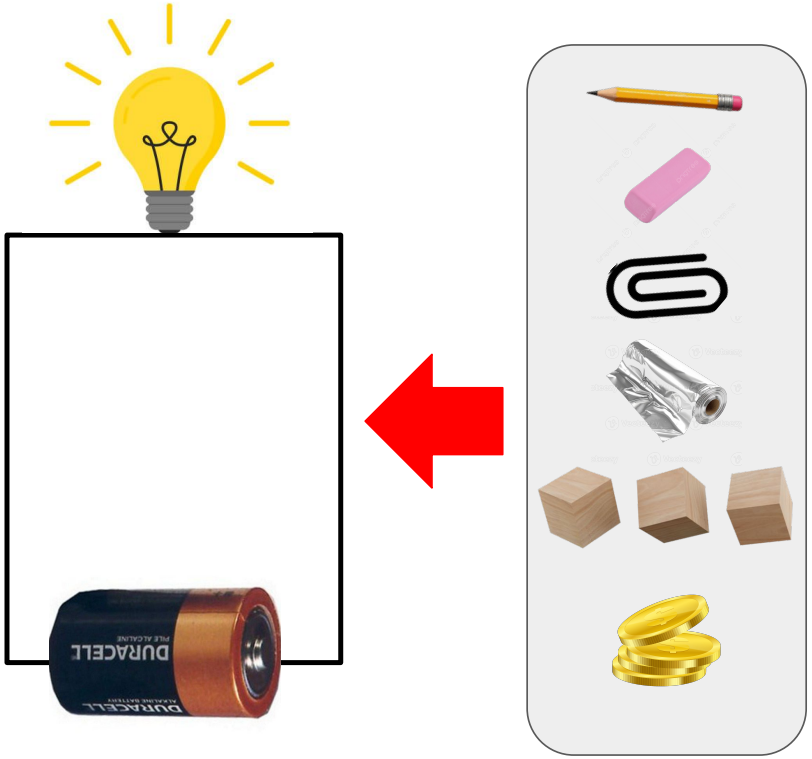
I will..

- Classify each material as conductor or insulator
- Explain the difference between a conductor vs insulator
- Rank materials from lowest to highest conductivity using data.

Review: Yesterday's Activity

- We tested different objects in a simple circuit with a battery and light bulb

- We observed whether the bulb **turned on, was dim or stayed off** for each material



Review

Conductor

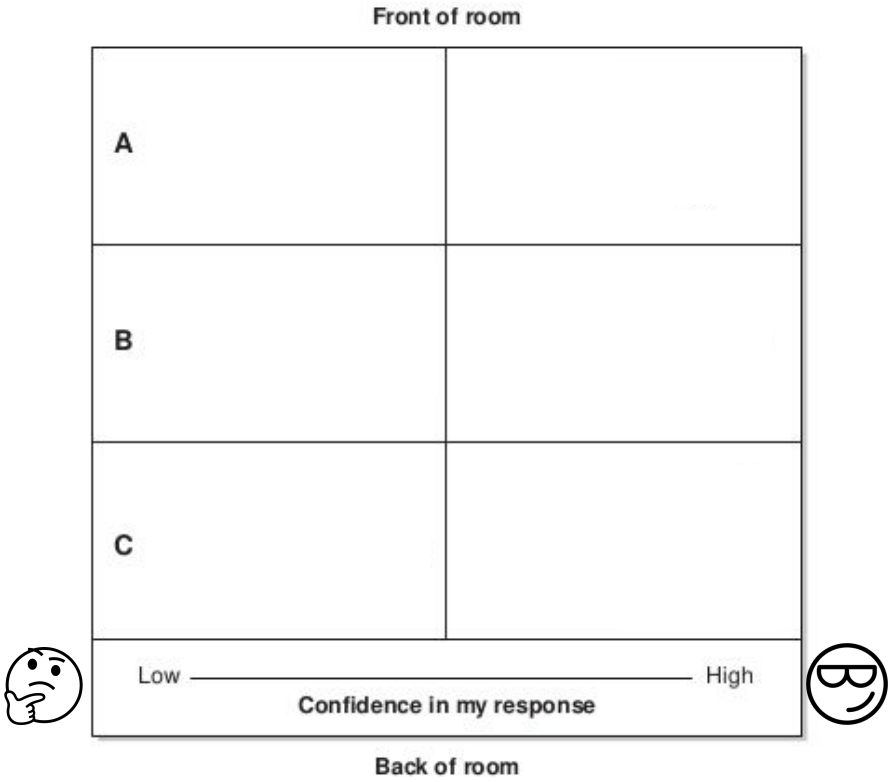
- Lets electric charge flow easily
- Electrons are **free to move**
- Usually metals (copper, aluminum, steel)

Insulator

- Does **not** let electric charge flow easily
- Electrons are **tightly held** and cannot move
- Rubber, plastic, wood, glass

Activity Time!!!

- Everyone will get a question
- Circle the answer **you** think works
- Stand in the spot that matches your answer



Question 1

What makes a material a good conductor?

A. It has no electrons

B. It has heavy atoms

C. It has tightly held electrons

D. It has free electrons that can move

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In conductors, the outer electrons are loose and can move freely. These moving electrons let electricity flow easily

Question 2

From our experiment yesterday, which is most likely an insulator?

A. Rubber band

B. Copper wire

C. Steel paperclip

D. Aluminum foil

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A rubber band is most likely an insulator because the bulb did not light showing it does not allow current to flow

Question 3

In the activity yesterday, the bulb shone brightly. What is the best conclusion?

A. The battery is empty

B. The bulb is broken

C. The material is an insulator

D. The material is a conductor

Question 3

In the activity yesterday, the bulb lights brightly. What is the best conclusion?

A. The battery is empty

B. The bulb is broken

C. The material is an insulator

D. The material is a conductor

If the bulb lights, electrons are flowing through the material. That means the material is a conductor

Question 4

Why are metals usually good conductors of electricity?

A. They have free electrons

B. Their atoms are very heavy

C. Their electrons are fixed in place

D. They have no electrons in the outer shell

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D. They have no electrons in the outer shell

Metals have “free electrons” that are not stuck to one atom. These electrons drift through the metal, carrying out current

Question 5

From our activity, which observation is the best evidence that a material is a good conductor?

A. Bulb is bright

B. Bulb flickers randomly

C. Bulb is off

D. Bulb is very dim

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From our activity, which observation is the best evidence that a material is a good conductor?

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The brighter the bulb, the more electrons are moving through the circuit. A bright bulb means the material conducts well

Information

Specific expectation

D2.2

Determine the conductivity of various materials by investigating their ability to hold or transfer electric charges

FACT

Human Scatterplot

BONUS

Bonus Question

Why are tool handles often coated in plastic?

A. Plastic creates more free electrons

B. Plastic lowers the tool’s mass number

C. Plastic conducts heat and electricity

D. Plastic is an insulator and improves safety

Bonus Question

Why are electrical tool handles often coated in plastic?

A. Plastic creates more free electrons

B. Plastic lowers the tool's mass number

C. Plastic conducts heat and electricity

D. Plastic is an insulator and improves safety

Plastic does not let electronics flow, so it blocks current.
This protects people from shocks when using metal tools